



Project Proposal: PrediCT

Building and Comparing Segmentation Strategies for Coronary Artery Calcium (CAC)

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1 Abstract

Accurate segmentation of coronary artery calcium (CAC) in non-contrast cardiac CT scans is essential for automated quantitative scoring and assessing the risk of Major Adverse Cardiovascular Events (MACE). However, this task is highly challenging due to the small, irregular, and sparse nature of calcified lesions, severe class imbalance, and frequent false positives from surrounding osseous structures such as ribs and vertebrae. Developing a robust, fully automated pipeline requires overcoming imaging noise and partial volume effects without suffering model collapse when processing scans without calcification. This project aims to build and systematically evaluate multiple machine learning segmentation strategies on the Stanford COCA dataset. A robust preprocessing pipeline will be implemented, including Hounsfield Unit (HU) windowing and strategic data augmentation. For the modeling phase, I will train and compare diverse architectural paradigms, including 2.5D U-Net variants, hybrid CNN–Transformer models, and domain-adaptation/self-supervised approaches, and benchmark their performance against nnU-Net, a strong and well-established baseline for medical image segmentation. The final deliverable will be a reproducible, open-source pipeline for preprocessing, model training, evaluation (using standard metrics such as the Dice coefficient), and automated Agatston scoring, establishing a foundation for scalable clinical modeling on large Stanford COCA datasets.

2 Literature Review

Medical image segmentation has been fundamentally advanced by the U-Net architecture [Ronneberger et al. \(2015\)](#), which employs a symmetric encoder-decoder design with skip connections to achieve precise localization, even with limited annotated data. Variants such as nnU-Net [Isensee et al. \(2021\)](#) have further improved performance by providing self-configuring pipelines that adapt to diverse imaging modalities and segmentation tasks. SegResNet [Liao et al. \(2026\)](#), a 3D residual CNN, extended this approach to volumetric data by leveraging residual connections and group normalization, improving local feature extraction in deep 3D networks. Complementary to these CNN-based approaches, 2.5D Multi-scale Feature Fusion Attention U-Net (MFFAU-Net) [Sun et al. \(2023\)](#) introduced an efficient pseudo-3D method by simultaneously processing adjacent slices and applying multi-scale attention to capture inter-slice contextual information with reduced computational cost.

Despite these advancements, conventional CNNs often struggle to capture long-range dependencies in 3D volumetric data due to their inherent spatial inductive biases. Hybrid CNN-Transformer models, such as TransUNet [Chen et al. \(2021\)](#), address this limitation by combining convolutional encoders for local feature extraction with Vision Transformer bottlenecks to model global self-attention. Extending this concept to fully volumetric data, nnFormer [Zhou et al. \(2023\)](#) integrates 3D transformers with interleaved convolutional layers, enabling robust segmentation of complex anatomical structures such as coronary arteries by modeling both local and long-range dependencies. Furthermore, self-supervised hybrid U-Nets [Soh and Rajapakse \(2025\)](#) have demonstrated the value of pretraining encoders on unlabeled scans using contrastive learning, which significantly improves generalization and segmentation performance when fine-tuned on small annotated datasets.

Robustness to domain shifts remains a critical challenge in medical imaging. Adversarial unsupervised domain adaptation techniques, such as those introduced by Kamnitsas et al. [Kamnitsas et al. \(2017\)](#), enable models to generalize across datasets with differing acquisition protocols, improving performance without extensive retraining.

In the context of Coronary Artery Calcium (CAC) scoring, deep learning methods have shown substantial clinical promise. Wang et al. [Wang et al. \(2024\)](#) performed a meta-analysis on automated CAC scoring, emphasizing the potential for rapid and accurate quantification. Eng et al. [Eng et al. \(2021\)](#) validated deep learning CAC scoring across multiple centers,

highlighting its reliability in heterogeneous populations. Ensemble approaches, such as the cascaded U-Net proposed by Alirr and Khalifa [Alirr and Khalifa \(2025\)](#), have proven effective in detecting subtle calcifications, while Hao et al. [Hao et al. \(2026\)](#) leveraged semi-supervised learning with scoring-driven consistency and negative sample boosting to address the common issue of data sparsity.

Clinical applicability is a key consideration in current research. Xu et al. [Xu et al. \(2025\)](#) demonstrated that deep learning-derived CAC scores from non-gated chest CTs are predictive of major adverse cardiovascular events (MACE) in patients with type 2 diabetes. Tang et al. [Tang et al. \(2025\)](#) reviewed the broader clinical impact of deep learning-based CAC quantification, emphasizing its potential for routine risk assessment. Furthermore, hybrid CNN-Transformer architectures [Alirr and Khalifa \(2025\)](#); [Jiang et al. \(2026\)](#) and self-supervised pretraining frameworks [Soh and Rajapakse \(2025\)](#) have been shown to improve segmentation precision and generalization, supporting the development of clinically robust Agatston scoring systems.

Collectively, these studies illustrate the evolution of medical image analysis: from CNN-based approaches to pseudo-3D and hybrid transformer-enhanced volumetric models, and finally to self-supervised frameworks capable of producing clinically actionable CAC scores. Building on this foundation, the present GSoC project will systematically evaluate these architectures on the Stanford COCA dataset, comparing traditional CNNs with hybrid, attention-based, and self-supervised models, bridging the gap between pixel-level segmentation performance and clinically meaningful calcium scoring.

3 ML4SCI Evaluation Task Results

3.1 Common Task 1: Dataset Preprocessing & Ground Truth Generation

- **Methodology:** A robust, automated volumetric data pipeline was implemented using **MONAI** and **PyTorch** to handle raw medical imagery:
 - **Pseudo-Labeling & Calcium Masking:** Addressed the lack of voxel-level manual annotations by deploying the TotalSegmentator tool to auto-generate anatomical ground truth masks for the heart. Simultaneously, a custom script mapped XML-based manual calcium point coordinates to the DICOM physical space using SimpleITK and OpenCV to generate dense 3D coronary calcium masks.
 - **Convex Hull Refinement (Solving the Calcium Paradox):** Initial experiments revealed that deep learning models were overfitting to the myocardium boundary and excluding epicardial calcifications. To solve this, a custom transform (**ConvexHullFill1d**) was engineered and applied slice-by-slice. This acted as a "shrink wrap" that naturally encompassed the coronary arteries and established a clinical safety buffer without adding excessive background.
 - **Preprocessing Strategy:** Applied uniform spatial resampling ($1.5 \times 1.5 \times 1.5mm^3$), strict Hounsfield Unit (HU) windowing clipped between -160 and 240 to perfectly isolate cardiovascular soft tissues while preserving calcium intensities, and dynamic spatial patch cropping using **RandCropByPosNegLabeld** to target positive heart regions.
 - **Data Augmentation:** Applied spatial augmentations including random affine transformations, flips, rotations, and Gaussian noise to mitigate overfitting.
- **Results Achieved:**
 - Successfully processed the 50-patient cohort, extracting valid image-mask pairs and establishing a strict dataset split yielding exactly **34 volumes for training, 8 for validation, and 8 for testing**.

- Conducted an Exploratory Data Analysis (EDA) that quantified the severe volumetric class imbalance: $\sim 75.8\%$ background, $\sim 24.0\%$ heart tissue, and a highly sparse $\sim \mathbf{0.04\%}$ **calcium**.
- Successfully implemented an efficient PyTorch DataLoader capable of dynamically applying the custom slice-wise convex hull refinement and augmentations on-the-fly to standardized volumetric batches.

3.2 Common Task 2: Heart Segmentation Model Training

- **Methodology:** Leveraged the preprocessed 3D volumes to train and evaluate segmentation networks specifically a Baseline 3D U-Net and **SegResNet** against a TotalSegmentator baseline.
 - **Handling Class Imbalance:** Utilized a **Dice Loss** function to optimize for spatial overlap rather than raw voxel accuracy, effectively navigating the extreme sparsity of the target structures. Models were optimized using Adam and a Cosine Annealing learning rate scheduler.
 - **Architecture Optimization:** Integrated residual blocks within the SegResNet encoder-decoder framework to extract deeper 3D volumetric context. By training on the convex hull refined labels, the network was forced to learn the critical clinical "shrink wrap" boundary.
 - **Diagnostic Metric:** Beyond standard anatomical overlap, a domain-specific custom metric (Calcium Inclusion Score) was calculated to ensure the predicted volume successfully encapsulated the true calcium voxels.
- **Results Achieved:** Evaluated on the 8 dedicated unseen test volumes using sliding window inference, focusing on Test Dice Score (accuracy), Inference Time (speed), and Calcium Inclusion (diagnostic viability). The comparative performance is summarized in Table 1 below.

Table 1: Performance Comparison of Heart Segmentation Models

Model	Test Dice Score	Inference Time (sec/vol)	Avg Calcium Included
TotalSegmentator	N/A	~ 33.55	100% (Ref)
Baseline 3D U-Net	0.8962	0.1119	99.18%
SegResNet	0.9413	0.0934	91.53%

- **Baseline 3D U-Net:** Achieved a strong Test Dice Score with an excellent **99.18%** calcium inclusion rate and a $305\times$ speedup over the baseline. However, its anatomical Dice score plateaued, indicating limitations in capturing complex structures.
- **SegResNet (Optimal):** Delivered the highest anatomical Test Dice Score of **0.9413** while maintaining an ultra-fast inference speed of **0.0934 seconds per volume** (a massive **$365\times$ speedup** over the traditional baseline). The convex hull refinement strategy ensured a robust calcium inclusion rate of **91.53%**, securing its utility as a scalable, real-time pipeline for downstream diagnostic scoring.

3.3 Code Repository

All preprocessing and segmentation code for the ML4SCI tasks is available at the following GitHub repository: <https://github.com/chayon20/PrediCT/blob/main/predict-projects.ipynb>

4 Detailed Methodology & Technical Approach

4.1 Data Acquisition and Preprocessing Pipeline

The foundation of this project relies on robust preprocessing of the Stanford COCA dataset. Non-contrast cardiac CT scans present unique noise profiles, severe class imbalance, and partial volume effects [Tang et al. \(2025\)](#). The pipeline, built on PyTorch and MONAI, will implement systematic intensity normalization. Specifically, I will apply Hounsfield Unit (HU) windowing strictly targeted at the attenuation range of calcium (e.g., > 130 HU). To address the frequent false positives generated by high-intensity osseous structures, I will develop a spatial masking or thresholding heuristic during the preprocessing phase to de-emphasize peripheral regions typically containing ribs and vertebrae, focusing the network’s attention on the cardiac bounding box.

4.2 Strategic Data Augmentation and Sampling

Coronary artery calcium (CAC) lesions are notoriously small, scattered, and subject to severe class imbalance compared to background tissue [Hao et al. \(2026\)](#). To mitigate overfitting and improve generalization, I will implement a rigorous, MONAI-driven augmentation pipeline utilizing elastic deformations, random cropping, affine translations, and Gaussian noise injection. Furthermore, I will employ a dynamic, patch-based sampling strategy during training. By intentionally oversampling patches containing minute calcifications and “hard negative” patches (e.g., those containing bone structures or dense artifacts), the model will be forced to learn discriminative features rather than collapsing to a majority-class prediction [Hao et al. \(2026\)](#).

4.3 Core Segmentation Architectures

To balance architectural efficiency with long-range contextual modeling, I will systematically train and compare six network topologies, progressing from optimized baselines to advanced hybrid and self-supervised models. These will include 2.5D U-Net variants, hybrid CNNTransformer architectures, and domain-adaptation/self-supervised models, and will be evaluated against nnU-Net, a strong and well-established baseline for medical image segmentation.

1. **nnU-Net (Gold Standard Baseline):** Acting as the foundational baseline, nnU-Net provides an auto-configuring framework that dynamically adapts its 2D, 3D, or cascaded U-Net pipeline based on dataset properties like resolution and class frequencies [Isensee et al. \(2021\)](#); [McConnell et al. \(2023\)](#). It ensures all subsequent advanced architectures are measured against a highly optimized state-of-the-art standard.(see Figure 1)

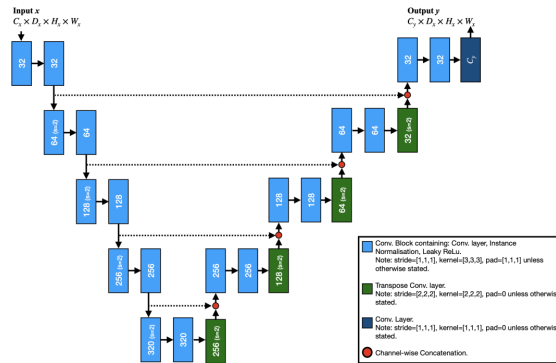


Figure 1: Architecture overview of the auto-configuring nnU-Net pipeline.

2. **SegResNet (Residual 3D CNN)**: Built within the MONAI framework, this 3D encoder-decoder model relies heavily on residual connections and group normalization [Liao et al. \(2026\)](#). It excels at local feature learning, preventing vanishing gradients in deep volumes, and provides a strong baseline for capturing precise local boundaries of target structures. (see Figure 2)

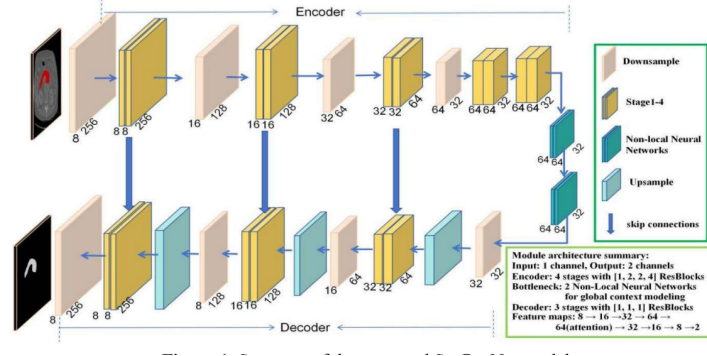


Figure 2: The 3D residual encoder-decoder structure of SegResNet.

3. **2.5D MFFAU-Net (Multi-scale Feature Fusion Attention U-Net)**: To balance computational efficiency with the need for volumetric context [Sun et al. \(2023\)](#), this model inputs multiple adjacent 2D slices simultaneously (e.g., $z - 1, z, z + 1$). This 2.5D approach enables the extraction of vital inter-slice spatial features along the z-axis, bypassing the extensive memory requirements associated with fully 3D convolutional networks (see Figure 3).

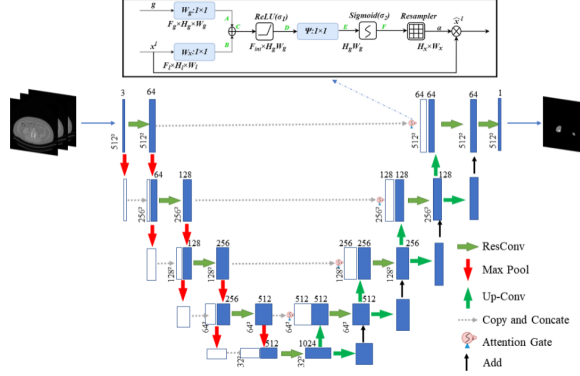


Figure 3: Feature extraction flow in a 2.5D MFFAU-Net.

4. **TransUNet (Hybrid CNN-Transformer)**: Combining the strengths of both paradigms, TransUNet uses a CNN encoder to extract high-resolution spatial feature maps, which are then passed through a Vision Transformer (ViT) bottleneck [Chen et al. \(2021\)](#). This intermediate hybrid model effectively learns global self-attention while retaining local spatial details. (see Figure 4)
5. **nnFormer (Full 3D Hybrid Transformer)**: Serving as the primary advanced model, nnFormer interleaves convolution and self-attention directly within a full 3D volume. By utilizing volume-based global self-attention, it can track long, winding arterial structures without breaking continuity and heavily reduces false positives by understanding the complete anatomical context [Zhou et al. \(2023\)](#). (see Figure 5)

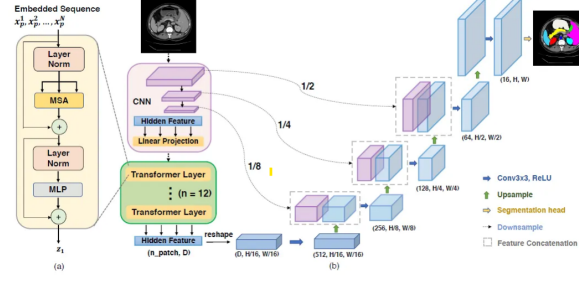


Figure 4: TransUNet architecture incorporating a CNN encoder and Vision Transformer bottleneck.

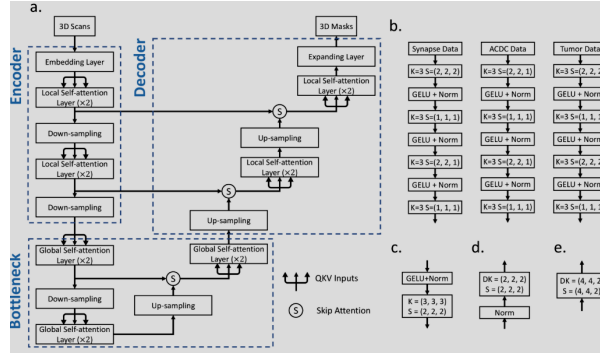


Figure 5: The nnFormer architecture utilizing 3D volumetric self-attention.

6. **Self-Supervised Hybrid U-Net:** Addressing the challenge of small, annotated medical datasets, this approach pretrains the encoder on unlabeled scans using contrastive learning [Soh and Rajapakse \(2025\)](#). Once the network learns the underlying human anatomy, it is fine-tuned on the labeled dataset, significantly improving data efficiency and generalization to unseen data. (see Figure 6)

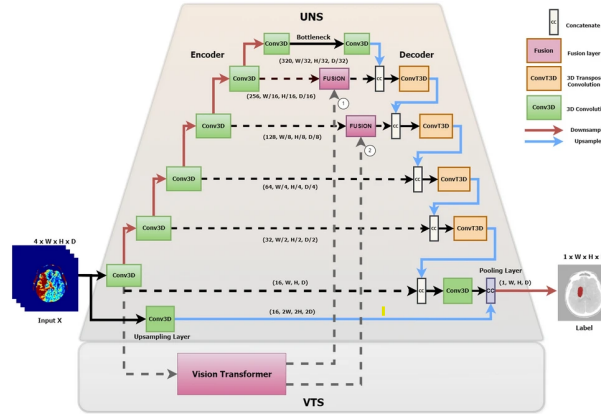


Figure 6: Pretraining and fine-tuning pipeline of the Self-Supervised U-Net.

Advanced Loss Functions: To address scans with zero calcium and high class imbalance, I will implement a dynamically weighted composite loss function, such as $L_{total} = \alpha L_{Dice} + \beta L_{Focal}$, allowing the model to heavily penalize false positives on healthy scans while maximizing overlap on segmented plaques [Hao et al. \(2026\)](#).

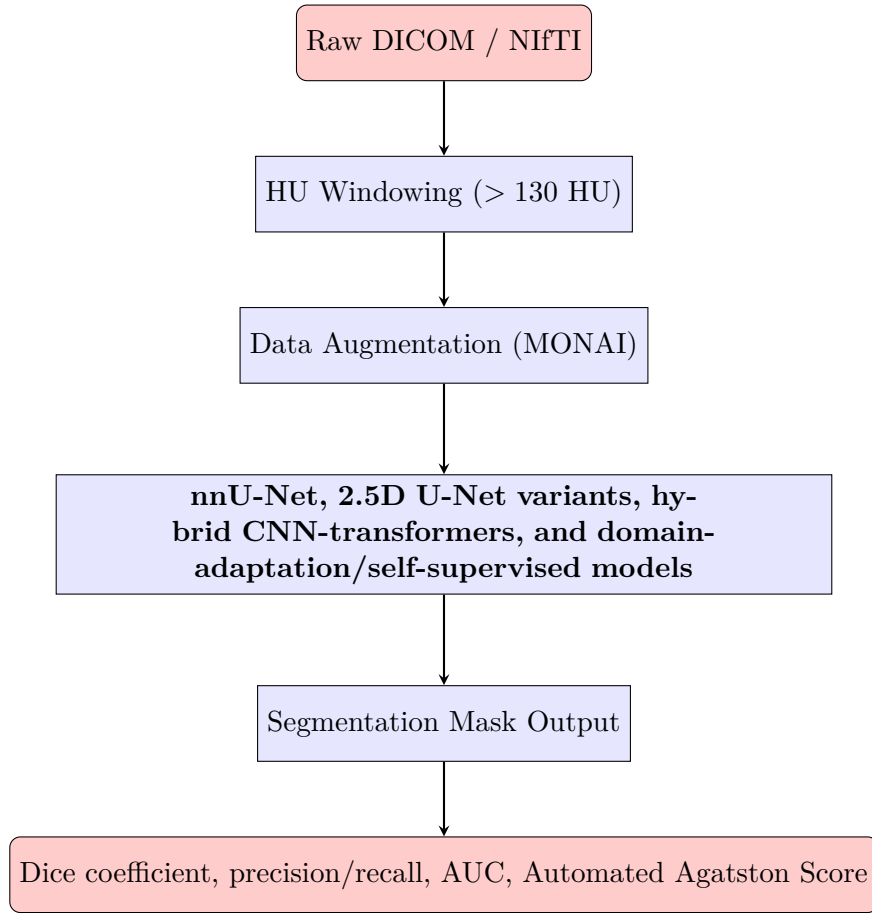


Figure 7: Proposed Segmentation and Scoring Pipeline Architecture.

4.4 Anatomical Localization and Multi-Task Learning

Beyond binary segmentation, clinical utility requires identifying the specific vascular territory of the plaque. Where dataset labels permit, I will implement an anatomically guided localization head [Alirr and Khalifa \(2025\)](#). This will be framed either as a multi-class segmentation task or a secondary classification branch that assigns segmented CAC masks to specific coronary arteries (Left Anterior Descending [LAD], Left Circumflex [LCX], or Right Coronary Artery [RCA]). This contextual awareness acts as an auxiliary learning signal, further refining the model’s spatial accuracy.

4.5 Automated Agatston Scoring and Clinical Translation

The ultimate test of a segmentation model in this domain is its downstream clinical validity and ability to predict Major Adverse Cardiovascular Events (MACE) [Xu et al. \(2025\)](#). I will write a dedicated Python script to automate the calculation of the Agatston score ($Score = Area \times DensityFactor$) directly from both the ground-truth annotations and the model’s predicted output masks [Eng et al. \(2021\)](#). This script will identify connected components in the predicted mask, calculate the maximum HU within each lesion to determine the density weight, and compute the total burden.

4.6 Benchmarking, Evaluation, and Optimization

Models will be subjected to rigorous, reproducible benchmarking [Wang et al. \(2024\)](#).

- **Spatial & Clinical Metrics:** Performance will be quantified using the Dice similarity coefficient, Intersection over Union (IoU), precision/recall curves, and Area Under the Curve (AUC). Crucially, I will perform a quantitative comparison of the computed Agatston scores (predicted vs. ground truth) using Pearson correlation and Bland-Altman analysis to prove clinical viability [Eng et al. \(2021\)](#); [Wang et al. \(2024\)](#).
- **Deployment Efficiency:** Drawing on my previous experience with PyTorch model quantization, I will explore **Quantization Aware Training (QAT)** and **static quantization**. This will reduce the memory footprint of the heavier hybrid models, ensuring the resulting pipeline is deployable for future clinical modeling on massive, Stanford COCA datasets.

5 Proposed Timeline

This project is structured over a 12-week coding period, aligning with the expected deliverables for the Machine Learning for Science PREDICT project.

- **Community Bonding Period:** Engage with mentors to finalize project scope and communication channels. **Thoroughly read the required research papers on different model architectures** (e.g., automated CAC scoring, nnU-Net, hybrid transformers) and the required PyTorch library documentation.
- **Phase 1 (Weeks 1–4) - Preprocessing & Baseline Models:**
 - *Weeks 1–2:* Finalize PyTorch and MONAI data loaders based on best practices from the literature review. Implement > 130 HU windowing, spatial masking heuristics to suppress osseous structures, and a robust data augmentation pipeline.
 - *Weeks 3–4:*
 - * Deploy and train **nnU-Net** as the gold-standard baseline.

- * Implement and train **SegResNet (3D CNN baseline)** for strong local feature learning comparison.
- * Build and train the **2.5D MFFAU-Net** to capture inter-slice contextual information efficiently.
- * Implement evaluation scripts for Dice, IoU, Precision/Recall, and AUC.
- *Deliverable:* Fully functional preprocessing pipeline and baseline models (nnU-Net, SegResNet, 2.5D U-Net) with initial benchmark results.
- **Phase 2 (Weeks 5–8) - Hybrid Architectures & Multi-Task Learning:**
 - *Weeks 5–6:*
 - * Implement and train **TransUNet (CNN–Transformer hybrid)** for global context modeling.
 - * Develop and train **nnFormer (full 3D Transformer-based model)** for volumetric self-attention.
 - * Introduce dynamic patch-based sampling and the composite loss function:
$$L_{total} = \alpha L_{Dice} + \beta L_{Focal}$$
to handle class imbalance and negative samples.
 - *Weeks 7–8:*
 - * Develop and integrate the **anatomical localization head** (LAD, LCX, RCA classification).
 - * Perform extensive hyperparameter tuning across all trained models.
 - * Conduct comparative evaluation between CNN, 2.5D, and Transformer-based architectures.
 - *Deliverable:* Hybrid models (TransUNet, nnFormer) and multi-task learning framework integrated. Midterm evaluation ready.
- **Phase 3 (Weeks 9–11) - Self-Supervision, Clinical Scoring & Optimization:**
 - *Weeks 9–10:*
 - * Implement **Self-Supervised Hybrid U-Net** using contrastive pretraining on Stanford COCA CT data.
 - * Fine-tune pretrained encoder on labeled CAC segmentation task.
 - * Develop the automated **Agatston scoring pipeline**.
 - * Perform clinical validation using Pearson correlation and Bland–Altman analysis.
 - *Week 11:*
 - * Apply model optimization techniques:
 - Quantization Aware Training (QAT)
 - Static quantization
 - * Benchmark inference speed, memory footprint, and deployment feasibility.
 - *Deliverable:* Self-supervised model, clinical scoring system, and optimized deployable models.
- **Phase 4 (Week 12) - Final Integration & Documentation:**
 - *Week 12:*
 - * Final comparative analysis across all six models:

- nnU-Net
- SegResNet
- 2.5D MFFAU-Net
- TransUNet
- nnFormer
- Self-Supervised U-Net
- * Refactor and document the complete codebase.
- * Prepare reproducible pipelines, README, and user guides.
- * Write and submit final GSoC report and present findings to the ML4Sci community.
- *Deliverable:* Final codebase, reproducible experiments, and complete project report.

6 About Me & Relevant Experience

I am currently pursuing a B.Sc. in Internet of Things & Robotics Engineering and work as a part-time Research Assistant focusing on Computer Vision, Deep Learning, and IoT.

I have a strong foundation in developing highly accurate deep learning models for both segmentation and classification using PyTorch. Recently, I developed a skin cancer detection system using the HAM10000 dataset, focusing on robust training pipelines and performance optimization. I have also engineered AI-driven IoT pipelines for agricultural applications, which led to a research paper titled “**An Explainable Quantized Transfer Learning Framework for Multiclass Rice Disease Detection with IoT-Based Soil Monitoring**”, currently under peer review in a **Q1 journal**.

In addition, I have hands-on experience in optimizing deep learning pipelines, including PyTorch model quantization and pruning techniques to improve inference efficiency. I also have a proven track record in academic manuscript writing and research documentation.

- **Languages:** C++, Python
- **Frameworks & Libraries:** PyTorch, OpenCV, MONAI, scikit-learn
- **Skills:** Computer Vision, Deep Learning Architecture, Model Quantization, Algorithm Design, Research Writing

My background in competitive programming, including back-to-back participation in the **ICPC Dhaka Regional contests (2024 and 2025)**, alongside active participation in **Kaggle competitions**, has equipped me with the rigorous problem-solving and optimization skills necessary to tackle algorithmic bottlenecks in complex tasks like 3D medical image processing.

7 Commitments & Availability

Hours per week: I commit to dedicating approximately 45-50 hours per week to this project during the official coding period. I am currently in my final semester at university, and the semester is primarily focused on internships and project work. Therefore, I will not have any significant time management conflicts and can fully concentrate on this project.

Time Zone & Communication: I am based in Dhaka, Bangladesh (UTC+6). I am highly flexible with my working hours and will be fully available to attend weekly sync meetings at times preferred by the mentors.

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